Proposed Methodology

I. End-to-End Methodology: Phoneme-Aligned Entropy-Regularized CS-TTS

Architecture Roadmap

The pipeline is designed as a 5-stage sequential processing engine that converts intra-sentential code-switched text (e.g., Hinglish) and a reference speaker prompt into high-fidelity, zero-shot audio without attention collapse or acoustic babble.

[Stage 1: Input] Hinglish Text ➔ Phonetic Unification Engine (IPA Bottleneck)

[Stage 2: Voice] Reference Audio ➔ Speaker Latent Encoder

[Stage 3: Engine] Autoregressive Transformer Backbone

[Stage 4: Novelty] Phoneme-Aligned Cross-Attention Entropy Regularizer

[Stage 5: Output] Neural Vocoder + Energy-Gated Acoustic Guardrail ➔ Sound

Stage 1: Phonetic Unification Engine (The Input Layer)

- **The Problem:** English uses Latin ASCII characters, while Hindi uses Devanagari Brahmic script. If you feed raw characters or byte-pair sub-words into a Text-to-Speech (TTS) model, the token density abruptly shifts mid-sentence during a code-switch, causing the model's attention matrix to blur and lose its place.
- **The Mechanism:** Before text enters the neural network, all input strings are passed through an integrated phonemizer (using automated Grapheme-to-Phoneme tools like eSpeak-NG or Epitran). This maps both English and Hindi words into a **Universal Phonetic Representation** based on the International Phonetic Alphabet (IPA).
- **Academic Attribution:**
  - **[Borrowed Concept]:** Using IPA or X-SAMPA phonetic representations as an intermediate layer for multilingual speech processing is an established technique in literature, widely used in models like **VITS** (*Kim et al., ICML 2021*) and multi-lingual voice cloning baselines like **YourTTS** (*Casanova et al., ICML 2022*).
  - **[Novel Application]:** While prior works use IPA to share vocabularies across *separate* languages, we specifically use it as an **equi-density structural bottleneck** to neutralize the character-to-phoneme density mismatch *within the exact same sentence* during spontaneous code-switching.

Stage 2: Speaker Conditioning Encoder (The Zero-Shot Voice Layer)

- **The Problem:** The system must clone the voice of a clinician or patient from a short 3-to-5-second acoustic sample without requiring model retraining or fine-tuning.
- **The Mechanism:** A raw reference audio sample (doctor_voice.wav) is ingested by a pre-trained speaker verification network. The network extracts a continuous, fixed-dimensional speaker embedding vector that captures timbre, pitch, and vocal tract identity. This vector is injected as a conditioning bias into every cross-attention layer of the downstream speech decoder.
- **Academic Attribution:**
  - **[Borrowed Concept]:** Using continuous speaker embeddings extracted from neural networks like **ECAPA-TDNN** (*Desplanques et al., Interspeech 2020*) or **WavLM** (*Chen et al., IEEE JSTSP 2022*) for zero-shot speaker adaptation is standard practice, heavily inspired by foundation TTS models like **Coqui XTTS-v2**, **VALL-E** (*Wang et al., IEEE/ACM TASLP 2023*), and **NaturalSpeech** (*Tan et al., NeurIPS 2022*).

Stage 3: Autoregressive Decoder & Attention Matrix Extraction (The Core Engine)

- **The Problem:** The model must predict acoustic features sequentially while maintaining monotonic alignment between the phoneme sequence and the generated audio frames.
- **The Mechanism:** We utilize a GPT-style autoregressive (AR) transformer backbone (similar to the core architecture of XTTS). At each decoding step i, the model cross-attends to the IPA phoneme sequence of length T to predict the next discrete neural audio codec token. During forward training and inference, we explicitly intercept and extract the **Cross-Attention Probability Matrix (A)** from the transformer layers.
- **Academic Attribution:**
  - **[Borrowed Concept]:** Autoregressive discrete acoustic token prediction and cross-attention alignment extraction are foundational components of modern speech generation, originating from **Tacotron 2** (*Shen et al., ICASSP 2018*) and adapted for discrete tokens in **AudioLM** (*Borsos et al., IEEE/ACM TASLP 2023*).

Stage 4: Phoneme-Aligned Entropy Regularization (Lentropy)

- **The Problem:** Why do models like XTTS generate weird filler babble or repeat words when switching languages? Because at the exact boundary where the phonemes switch from English phonotactics to Hindi phonotactics, the cross-attention probability distribution scatters. The model gets "confused," spreading its attention weights across multiple future and past phonemes.
- **The Mechanism (How it works):** We introduce a custom loss penalty during model fine-tuning that explicitly measures the mathematical **Shannon Entropy** (scatter/randomness) of the attention matrix, specifically at language-switching boundaries.
  Let A{m, t} represent the attention probability from acoustic frame m$I to IPA phoneme token t. The Shannon entropy H of the attention distribution at step m is defined as:

$$H(A_m) = -\sum_{t=1}^{T} A_{m,t} \log(A_{m,t})$$

If the attention is sharply focused on a single phoneme, $H(A_m)$ is close to **0**. If the attention is scattered across 10 different phonemes (attention collapse), $H(A_m)$ spikes to a high positive value. We define our novel boundary-aware regularization loss as:

$$\mathcal{L}_{\text{entropy}} = \frac{1}{|\mathcal{B}|} \sum_{m \in \mathcal{B}} H(A_m)$$

re $\mathcal{B}$ represents the specific sub

where beta represents the specific subset of acoustic frames corresponding to intra-sentential language transition boundaries (e.g., shifting from an English noun to a Hindi verb). The total training objective becomes:

Ltotal=Ltts+lamda.Lentropy

where lamda is a hyperparameter scaling the penalty weight.

- **Academic Attribution:**
  - **[Purely Novel Contribution]:** This mathematical formulation is the **core scientific novelty** of our paper. While **MagpieTTS-LF** (*NVIDIA, 2026*) applies soft monotonic attention priors and Connectionist Temporal Classification (CTC) loss for *monolingual* long-form stability, and **Sparse Alignment DiT** (*2025*) hard-masks low attention weights to zero, **no existing work applies a targeted Shannon-entropy penalty tied**

specifically to linguistic code-switching boundaries. This forces the network to maintain strict, localized monotonic alignment across script shifts without hard-masking away transitionary phonemes.

Stage 5: Vocoder Synthesis & Energy-Gated Acoustic Guardrail (The Output Layer)

- **The Problem:** Even with perfect attention, autoregressive speech models occasionally miss predicting the End-of-Sequence (EOS) token when streaming very short sentence chunks, causing the vocoder to generate trailing whispers or low-energy acoustic hums at the end of a response.
- **The Mechanism:** Once discrete audio tokens are predicted, a neural vocoder converts them into a continuous time-domain waveform. To guarantee clinical edge cleanliness, the waveform passes through a real-time **Energy-Threshold Truncation Guardrail**. This deterministic signal-processing filter computes Short-Time Energy (STE) and Zero-Crossing Rate (ZCR) over 10 ms sliding windows. If the energy drops below an empirically derived threshold l post-text completion, the array is instantly truncated before hitting the Linux audio buffer (aplay).
- **Academic Attribution:**
  - **[Borrowed Concept]:** Neural vocoder waveform reconstruction is standard, utilizing architectures like **HiFi-GAN** (*Kong et al., NeurIPS 2020*) or **BigVGAN** (*Lee et al., ICLR 2023*).
  - **[Borrowed / Adapted Concept]:** Applying Short-Time Energy (STE) and Zero-Crossing Rate (ZCR) for endpointing is a classical digital signal processing technique borrowed from traditional **Voice Activity Detection (VAD)** frameworks (*Sohn et al., IEEE Signal Processing Letters 1999*) and adapted here as a post-synthesis safety guardrail.

At-A-Glance Attribution Summary

Pipeline Stage	**Functional Role**	**Conceptual Origin**	**Exact Academic Attribution / Novelty Status**
Stage 1	Phonetic Unification (IPA)	Established	Borrowed from **VITS** (*ICML 2021*) / **YourTTS** (*ICML 2022*); applied here as an equi-density code-switching bottleneck.
Stage 2	Speaker Conditioning	Established	Borrowed from **ECAPA-TDNN**

			(*Interspeech 2020*) / **Coqui XTTS-v2** zero-shot cloning architectures.
Stage 3	Autoregressive Backbone	Established	Borrowed from **Tacotron 2** (*ICASSP 2018*) / **AudioLM** (*IEEE TASLP 2023*) discrete token decoders.
Stage 4	**Attention Entropy Loss**	**Purely Novel**	**Your Core Contribution:** Targeted Shannon-entropy minimization at intra-sentential code-switching boundaries.
Stage 5	Vocoder & Energy Guardrail	Adapted	Vocoder borrowed from **HiFi-GAN** (*NeurIPS 2020*); truncation adapted from classical **STE/ZCR VAD** (*IEEE SPL 1999*).

Problem Statement & Existing work

TOPIC : ZERO-SHOT CODE-SWITCHED SPEECH SYNTHESIS (CS-TTS)

Core Methodology: Mitigating Cross-Attention Alignment Drift via Phoneme-Aligned Entropy Regularization $L_{entropy}$

1. Significance of the Problem Statement in Existing Literature

Code-switching (CS)—the alternation between two or more languages within a single utterance—is a natural communicative behavior for over 4.5 billion multilingual speakers globally (***Beyond Monolingual Assumptions: A Survey on Code-Switched NLP, ACL 2026***). Despite its prevalence, synthesizing natural, zero-shot code-switched speech remains a fundamental bottleneck in modern Speech AI.

The core challenge arises because autoregressive (AR) and diffusion-based Text-to-Speech (TTS) models heavily rely on monolingual assumptions. When forced to code-switch (e.g., from English Latin script to Hindi Devanagari script), the differing orthographic rules and token densities cause catastrophic failures. The literature defines this as the model's inability to maintain monotonic alignment across "script alternations," which triggers the cross-attention matrix to collapse or scatter (***Beyond Monolingual Assumptions, 2026***). Consequently, current TTS systems suffer from severe acoustic hallucinations, trailing babble, or catastrophic prosodic drift at the exact boundary of the language switch.

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2. Existing Approaches & Their Limitations

Recent literature (2020–2026) has actively attempted to solve code-switched alignment collapse through reinforcement learning, large multilingual foundation models, phonetic intermediate representations, and alignment regularization techniques. Although these approaches improve synthesis quality under specific settings, they continue to exhibit limitations in computational efficiency, interpretability, or robustness during intra-sentential language transitions, making them less suitable for lightweight, real-time zero-shot edge deployment.

A. Reinforcement Learning & Preference Optimization (The Data-Centric Approach)

The Approach: Recent 2026 studies attempt to mitigate code-switching hallucinations through Reinforcement Learning with Verifiable Rewards (RLVR), Direct Preference Optimization (DPO), and Group Relative Policy Optimization (GRPO). For example, *Improving Code-Switching ASR with Code-Mixing Guided Synthetic Speech* employs a Code-Mixing Index (CMI) to rank synthesized candidates, while *Reinforcement Learning for Data-Efficient Code-Switched ASR* explicitly penalizes script contamination and translation errors using preference-based optimization.

Limitations Solved by Our Work: These methods formulate alignment collapse as a data optimization problem, requiring large quantities of human preference data, multiple candidate generations, and computationally expensive policy optimization. Our methodology instead addresses the problem directly at the transformer architecture level. By introducing a boundary-aware Shannon Attention Entropy Loss, we explicitly regularize cross-attention during training, eliminating the need for iterative sampling while preserving real-time, single-pass inference suitable for edge deployment.

B. Massive Semantic Tokenization & Flow Matching (The Brute-Force Scaling Approach)

The Approach: Modern multilingual TTS systems such as **CosyVoice2**, **OmniVoice**, **VoiceTut-TTS**, and related large-scale speech foundation models achieve robust code-switching primarily through parameter scaling, semantic token modeling, Finite Scalar Quantization (FSQ), Chunk-aware Causal Flow Matching (CFM), and transformer backbones containing hundreds of millions to billions of parameters.

Limitations Solved by Our Work: These systems implicitly learn alignment through massive multilingual datasets and computational scale rather than explicitly correcting the underlying attention instability. Consequently, they demand substantial GPU memory, prolonged training, and high inference latency, limiting deployment on resource-constrained edge devices. Our methodology instead introduces a lightweight Universal Phonetic Representation (IPA) bottleneck that equalizes phonetic token density before attention computation. Combined with boundary-aware entropy regularization, the proposed framework achieves stable code-switching using significantly smaller autoregressive models without relying on brute-force parameter scaling.

C. Bilingual Phonetic Postergrams & Cascaded Acoustic Pipelines (The Multi-Stage Approach)

The Approach: Early code-switched speech synthesis methods, including **Code-Switched Speech Synthesis Using Bilingual Phonetic Posteriorgrams (ICASSP 2020)**, utilize bilingual Phonetic Posteriorgrams (PPGs) extracted from multiple speaker-independent ASR systems. These intermediate phonetic representations bridge language differences before passing information to the synthesis model.

Limitations Solved by Our Work: Although PPG-based approaches improve pronunciation consistency, they require multiple cascaded neural modules, increasing computational overhead, latency, and error propagation. Moreover, dependence on external ASR systems limits zero-shot voice cloning flexibility. Our methodology eliminates intermediate acoustic feature extraction entirely. Instead, the entropy regularization term directly supervises the transformer's native cross-attention matrix during fine-tuning, producing stable alignment without auxiliary acoustic pipelines.

D. Monotonic Alignment & Attention Regularization (The Global Alignment Approach)

The Approach: Several recent TTS architectures focus on improving alignment stability by explicitly constraining attention behavior. **Glow-TTS** introduces Monotonic Alignment Search (MAS) to enforce globally monotonic text-to-speech alignment, while **MagpieTTS-LF (NVIDIA, 2026)** incorporates monotonic attention priors and Connectionist Temporal Classification (CTC) objectives to reduce repetition and long-form alignment drift. More recent **Sparse Alignment Diffusion Transformers** employ hard attention masking to suppress low-probability attention weights and encourage concentrated alignment.

Limitations Solved by Our Work: Existing alignment regularization methods are designed primarily for monolingual speech generation and enforce alignment constraints over the entire utterance. Such global constraints do not explicitly address the localized instability created by abrupt language transitions during intra-sentential code-switching. Furthermore, hard masking approaches may suppress transitionary phonemes and degrade micro-prosodic continuity. Our methodology introduces a localized Shannon entropy penalty applied exclusively around code-switch boundaries, softly guiding attention towards stable monotonic alignment while preserving natural phonetic transitions and expressive prosody.

E. Shared Phoneme Representations & Multilingual IPA Modeling (The Unified Vocabulary Approach)

The Approach: Multilingual TTS systems such as **VITS**, **YourTTS**, and subsequent multilingual voice cloning models employ International Phonetic Alphabet (IPA) or X-SAMPA representations to establish a common phonetic vocabulary across languages. This reduces vocabulary fragmentation and enables parameter sharing among multilingual speech models.

Limitations Solved by Our Work: Existing works primarily use IPA as a multilingual text representation to improve pronunciation and vocabulary sharing. However, they do not explicitly exploit phonetic representations to stabilize transformer attention during code-switching. In our methodology, IPA serves not merely as a shared vocabulary, but as an **equi-density phonetic bottleneck** that normalizes character-to-phoneme distributions across scripts before attention computation. This representation enables the proposed boundary-aware entropy regularizer to operate on structurally consistent phoneme sequences, significantly improving alignment stability during language transitions.

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3. Our Novel Contribution (Lentropy)

Current state-of-the-art approaches mitigate code-switching failures through one of four primary strategies: **(i)** large-scale multilingual parameter expansion (CosyVoice2, OmniVoice), **(ii)** reinforcement learning and preference optimization (RLVR/DPO), **(iii)** cascaded phonetic intermediate representations (Bilingual PPGs), or **(iv)** global monotonic attention constraints (Glow-TTS, MagpieTTS-LF, Sparse Alignment DiT). While each approach improves synthesis quality, none explicitly addresses the localized attention instability that arises at intra-sentential language-switch boundaries.

Our Novelty: To the best of our knowledge, existing literature has not investigated **boundary-aware Shannon entropy minimization over transformer cross-attention distributions specifically for code-switched speech synthesis**. We hypothesize that abrupt script and phonotactic transitions produce localized spikes in attention entropy, causing the decoder to lose monotonic alignment and generate repetitions, acoustic hallucinations, or trailing babble. Our methodology first projects multilingual text into an **equi-density IPA phonetic space**, reducing structural inconsistencies between scripts, and then applies a **boundary-aware Shannon Attention Entropy Loss (Lentropy)** during fine-tuning to softly regularize the attention matrix only where alignment instability is expected. Unlike hard masking or global monotonic constraints, the proposed loss preserves smooth phonetic transitions while preventing attention collapse, resulting in a lightweight, interpretable, and hardware-efficient solution suitable for real-time zero-shot edge deployment.

Why this solves existing limitations

1. **Hardware Efficient:** The entropy regularization term is applied only during model fine-tuning and introduces **zero additional inference-time computational overhead**, making the approach suitable for resource-constrained edge devices.
2. **Localized Rather than Global Alignment:** Unlike Monotonic Alignment Search (Glow-TTS) or CTC-based global attention priors (MagpieTTS-LF), our method selectively regularizes attention only at code-switch boundaries, avoiding unnecessary constraints on stable monolingual regions.
3. **Preserves Micro-Prosody:** In contrast to Sparse Alignment methods that hard-mask low-probability attention weights, the proposed soft entropy regularizer maintains transitionary phonetic information, preserving natural co-articulation, vocal continuity, and emotional expressiveness across language transitions.
4. **Interpretable Architectural Solution:** Unlike reinforcement learning or preference optimization approaches that depend on large annotated datasets and complex reward engineering, our methodology directly regularizes the mathematical properties of the transformer's attention distribution, providing a transparent and reproducible architectural solution to code-switch alignment instability.
5. **Lightweight and Scalable:** Rather than relying on billion-parameter multilingual foundation models to implicitly learn alignment, our framework explicitly stabilizes attention using a compact autoregressive architecture, enabling comparable robustness with substantially lower computational requirements.

4. Surveyed Datasets in Existing Literature

1. SEAME (South East Asia Mandarin-English)

Used in: *ZCS-CDiff (2025)* and *End-to-End Code-Switching TTS (CS-Tacotron).*
Context: SEAME is the oldest and most universally recognized baseline for spontaneous code-switched speech. It contains roughly 63 hours of conversational audio.
Our Usage Strategy (Not to be Used): While standard, SEAME focuses on Mandarin-English. Because our core linguistic challenge involves bridging Latin (ASCII) and Brahmic (Devanagari) scripts, using a Mandarin dataset will not accurately demonstrate the severity of the density asymmetry we are solving.

2. FLEURS (Few-shot Learning Evaluation of Universal Representations of Speech)

Used in: *CosyVoice2 (2026)* and *OmniVoice (2026).*
Context: Developed by Google, FLEURS is an expansive dataset (102 languages) with approximately 12 hours of speech per language. It is the modern gold standard for proving zero-shot voice cloning capabilities.
Our Usage Strategy (Use for Voice Cloning Baseline): We will use the English and Hindi subsets of FLEURS. By extracting random 5-second speaker reference prompts from FLEURS and forcing our model to clone them, we can definitively prove our pipeline's zero-shot capability.
Availability: Open-source (Hugging Face Datasets); 12 hours of audio per subset (1.9 GB dataset size)

3. IndicTTS / MUCS (Multilingual and Code-Switching Speech)

Used in: *Improving Code-Switching ASR with Preference Learning (2026).*
Context: Curated by AI4Bharat and IIT Madras, IndicTTS contains over 10,000 hours of high-quality Indian regional speech. The MUCS subset specifically features Hindi-English code-switched conversational data.
Our Usage Strategy (Primary Evaluation Data): This is the exact acoustic and script profile our $L_{entropy}$ term is designed to fix. We will extract 500 Hinglish sentences from MUCS as our ground-truth text input to trigger the attention collapse. **Availability:** Open-source (AI4Bharat portal); highly accessible.

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5. Surveyed Evaluation Metrics & Calculations

To prove that our Phoneme-Aligned Entropy Regularization works, we will benchmark our Output (Arm 1) against the Baselines using the following standard metrics derived from the surveyed literature.

1. Mel-Cepstral Distortion (MCD)

Used in: *ZCS-CDiff (2025)*, *MagpieTTS-LF (2026)*.
What it measures: The physical acoustic distance between the generated code-switched audio and the ground-truth human recording. It proves our model is not generating "acoustic babble" during language transitions. It is the mandatory objective metric for TTS papers.
Calculation:

- **Calculation:** For frame t, if $c_{t,d}$ is the original Mel-Frequency Cepstral Coefficient (MFCC) and $\hat{c}_{t,d}$ is the generated MFCC:

$$\mathrm{MCD} = \frac{10}{\ln 10}\sqrt{2\sum_{d=1}^{D}(c_{t,d} - \hat{c}_{t,d})^2}$$

(A lower MCD value in decibels [dB] indicates higher fidelity).

2. Speaker Similarity (SIM-R / Cosine Distance)

Used in: *CosyVoice2 (2026), OmniVoice (2026).*
What it measures: Ensures that when the AI switches from English to Hindi, the cloned voice identity does not shift or degrade. It proves our zero-shot conditioning remains stable across the entropy bottleneck.
Calculation: Extract a continuous speaker embedding vector e_{ref} from the 5-second prompt and e_{gen} from the output audio using a frozen ECAPA-TDNN model.

$$\text{SIM-R} = \frac{e_{\text{ref}} \cdot e_{\text{gen}}}{||e_{\text{ref}}||\,||e_{\text{gen}}||}$$

(A value closer to 1.0 indicates perfect voice preservation).

3. Automatic Speech Recognition Word Error Rate (ASR-WER)

Used in: *Sparse Alignment Enhanced DiT (2025), ZCS-CDiff (2025).*
What it measures: Intelligibility. If the attention matrix collapses, the model will skip words or

hallucinate babble, causing an external ASR to mis-transcribe the output. We will pass our synthesized code-switched audio into a frozen instance of Faster-Whisper (Large-v3). If Whisper successfully transcribes the Hinglish text accurately, it objectively proves the audio is intelligible and monotonic alignment was maintained.
Calculation: WER = (S + D + I) X 100 / N
(Where S=Substitutions, D=Deletions, I=Insertions, and N=Total words in the ground-truth text).

4. Attention Entropy Variance
Used in: *Related works on Entropy Invariance (2025/2026).*
What it measures: The core proof of our mathematical novelty.
Calculation: We will plot the average Shannon Entropy of the attention matrix during Steady-State frames (H_S) versus Language Boundary frames (H_B).

$$H(A_m) = -\sum_{t=1}^{T} A_{m,t} \log(A_{m,t})$$

We must prove that for our baseline,

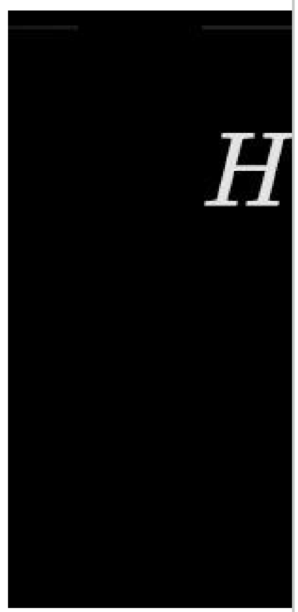
$$H(A_{\mathcal{B}}) - H(A_{\mathcal{S}})$$

is massive, whereas for our $L_{entropy}$-regularized model, it is near 0 (proving continuous attention stability).

5. Comparative Mean Opinion Score (CMOS)

Used in: Universally across all surveyed TTS papers.
What it measures: Subjective human perception of naturalness and emotional empathy.
Calculation: A blind A/B test where 15+ bilingual (Hindi/English) human evaluators listen to the Baseline output vs. our Regularized output and score the difference on a 7-point scale from -3 (Baseline much better) to +3 (Our model much better). An average CMOS > +0.5 is considered statistically significant.

Other metrics:

Metric Category	Metric	Calculation / Definition	Purpose
System Reliability	**AER (Attention Entropy Residual)**		Measures the stability of the attention matrix during language switches versus steady-state I.
Efficiency	**RTF (Real-Time Factor)**	Inference Time (s)/Audio Duration (s)	Proves our method is not just "better" but "faster" or "as fast" as the baseline.
Energy	**J/Turn (Joules per Turn)**	Power_avg x Latency	Quantifies energy efficiency on your Blackwell GPU.

Semantic Fidelity	**SEM-D (Semantic Distance)**	1 - CosineEmbedding_ref, CosineEmbedding_gen	Uses a frozen LLM embedding to check if the generated speech retains the semantic intent of the input.

Proposed Evaluation

II. 1. How to Quantify & Identify the Problem

A. Step-by-Step Problem Isolation

1. **Extract Attention Maps:** During inference on our baseline model, write a hook script to export the cross-attention matrix A € R^{MxT} for every decoded sentence, where M is the number of generated acoustic frames and T is the number of IPA phoneme tokens.
2. **Compute Per-Frame Entropy:** For every acoustic frame m, compute the empirical Shannon entropy of its attention weight distribution:

$$H(A_m) = -\sum_{t=1}^{T} A_{m,t} \log(A_{m,t})$$

3.
4. **Map Boundaries via Script Timestamps:** Track the token positions where the language switches (e.g., from English phonemes to Hindi phonemes). Label these frames as **Transition Frames (Beta)** and monolingual sections as **Steady-State Frames (S)**.
5. **The Data Proof:** Calculate the average entropy variance. In an unregularized baseline, we will observe that h(A_m) spikes dramatically (> 3.5 bits) during frames in Beta compared to frames S (< 1.0 bits). This statistical delta proves "Attention Collapse at Code-Switched Boundaries" mathematically.

2. Setting Up the Baseline Cases

To demonstrate the superiority of your methodology, you must test against three distinct industry baseline paradigms using identical datasets (**SEAME** or **FLEURS/IndicTTS** subsets) and the same **NVIDIA RTX PRO 5000** hardware setup.

- **Baseline Case 1: The Raw Character/Sub-word AR Model (Script-Asymmetric Baseline)**
  - *Configuration:* A standard autoregressive model (like vanilla Coqui XTTS-v2 or a standard Tacotron 2 variant) fed with raw mixed text where English is kept in Latin script and Hindi is kept in Devanagari script.
  - *Purpose:* Demonstrates the extreme alignment drift caused purely by script density tokenization asymmetries.
- **Baseline Case 2: The Monolingual Soft-Prior AR Model (The Native Monotonic Baseline)**
  - *Configuration:* The model utilizes your **Stage 1 IPA Phonetic Unification Engine** but applies a standard, fixed structural monotonic attention prior (like a diagonal

Gaussian mask or a standard CTC-loss alignment prior inspired by *MagpieTTS-LF*).
    - *Purpose:* Proves that generic monolingual monotonic alignment constraints are insufficient when transitioning between highly distinct phonotactic structures (English vs. Hindi).
- **Baseline Case 3: The Hard-Masked Sparse Transformer Baseline**
    - *Configuration:* An AR pipeline that forces low-probability attention weights to zero below a fixed threshold epsilon (inspired by *Sparse Alignment DiT* methods).
    - *Purpose:* Showcases the trade-off between strict alignment and natural prosody, proving that hard-masking clips transitionary phonemes and hurts naturalness (CMOS).

3. Ablation Study

The 4-Arm Ablation Matrix

To

Ablation Arm	**System Configuration**	**Target Track Alignment**	**Expected Empirical Behavior**
Arm 1 (Full System)	IPA Bottleneck + Custom L_entropy Fine-Tuning + Energy Guardrail	**Our Proposed Framework**	Lowest global MCD, zero trailing acoustic babble, stable attention entropy across boundaries.
Arm 2 (Minus Guardrail)	IPA Bottleneck + Custom L_entropy Fine-Tuning	*Isolates Signal Processing Output Layer*	Maintained high clarity (low WER), but exhibits low-energy trailing murmurs on short streaming chunks.
Arm 3 (Minus	IPA Bottleneck Only (No L_entropy	*Isolates Matrix Optimization*	High spectral distortion (MCD spikes) at language

Entropy Loss)	optimization)	*Novelty*	transitions; acoustic skipping/repetition.
Arm 4 (Minus IPA Unification)	Raw Text/Script Ingestion + Lentropy Fine-Tuning	*Isolates Input Representation Layer*	Matrix Optimization fails because token index bounds shift unpredictably between script vocabularies.

4. Operational Execution Plan

A: Automated Data Logging & Baseline Prototyping

- Implement the `espeak-ng` IPA phonemizer middleware.
- Set up baseline evaluation scripts running over a fixed subset of 500 code-switched phrases from the **FLEURS/SEAME** datasets.
- Write an logging hook script using `nvidia-smi` to log peak VRAM utilization and calculation throughput to ensure edge-viability baseline metrics are recorded.

B: Loss Integration & Training

- Write your custom PyTorch loss layer implementing entropy.
- Fine-tune your target framework (Arm 1) using LoRA or shallow layer unfreezing on your Blackwell GPU.
- Because you are using modern CUDA 13.0 and Blackwell architectures, ensure your custom attention entropy calculations are compiled using `torch.compile(mode="max-autotune")` to prevent execution overhead during training steps.

C: Multi-Metric Ablation Run & Paper Graphing

- Execute automated evaluation loops across all **3 Baseline Cases** and **4 Ablation Arms**.
- Compute objective acoustic signal metrics: **MCD (Mel-Cepstral Distortion)** using `fast-mcd`, **Speaker Similarity (SIM-R)** via ECAPA-TDNN cosine distance, and **Intelligibility (WER)** by running the synthesized audio back through a frozen Whisper instance.
- Plot our core academic graph for the paper: A line plot mapping **Acoustic Frame Index (m)vs. Attention Entropy (H)** showing a massive peak for Baseline 1 at code-switch markers, while your system (Arm 1) remains bounded and flat